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\documentclass[11pt]{article}

% Use Helvetica font.
%\usepackage{helvet}

%\usepackage[T1]{fontenc}
%\usepackage[sc]{mathpazo}
\usepackage{color}
\usepackage{amsmath,amsthm,amssymb,multirow,paralist}
%\renewcommand{\familydefault}{\sfdefault}

% Use 1/2-inch margins.
\usepackage[margin=0.8in]{geometry}
\usepackage{hyperref}
\usepackage{float}
\setlength{\parindent}{2em}
\setlength{\parskip}{\baselineskip}

\begin{document}

\begin{center}
{\Large \textbf{Sleep Efficiency Classification Using Machine Learning}

\vspace{10pt}

Project Midterm Report}

\vspace{10pt}

\textit{Team Members: Evan Litzer, Eric Huang, Madison Vosburg}
\end{center}

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\begin{abstract}
This project implements and compares three different machine learning models discussed in
class: linear perceptron, logistic regression, and support vector models. We are using a dataset
that contains information on sleep quality and lifestyle habits. Our task is to implement all of the
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above models using this dataset, compare their accuracies, and identify which features each model cares most about. This will also help us understand which factors contribute the most to quality sleep.
\end{abstract}

\section{Introduction}

\subsection{Data Set}
We are using the Sleep and Lifestyle Health dataset found on Kaggle. This is a CSV dataset constructed
with 1000 rows and 15 columns, where each column represents a different feature that factors into one’s
sleep length and quality. Factors include lifestyle choices such as time of sleep and wake, exercise, caffeine, smoking, and alcohol. This is a synthetic dataset that simulates the effects of different habits and health variables on sleep.

\subsection{Machine Learning Models}
\subsubsection*{1. Linear Perceptron}
The linear perceptron is a simple linear classifier that attempts to find a separating hyperplane between the two classes. It works well when the data is linearly separable. This model will serve as a baseline for comparison.

\subsubsection*{2. Logistic Regression}
Logistic regression is also a linear classifier, but produces probability outputs for each class. It is generally more stable than linear perceptron and performs better when classes overlap. It also allows us to interpret how individual features contribute to predictions.

\subsubsection*{3. Support Vector Machine (SVM)}
The SVM finds the optimal separating hyperplane by maximizing the margin between classes. SVM often performs better than other linear models because it focuses on the most important data points near the decision boundary. We expect SVM to achieve the best performance among the 3 models.

\subsection{Expected Outcomes}
We expect to observe relationships between lifestyle factors and sleep efficiency. For example, we predict
that higher caffeine consumption, higher alcohol intake, smoking status, and increased awakenings may
be associated with poor sleep efficiency. In contrast, regular exercise and balanced sleep composition may
be associated with good sleep efficiency.
\\\\
In terms of model performance, we expect:
\begin{itemize}

\item The linear perceptron to perform the worst if the data is not perfectly linearly separable.
\item Logistic regression to perform better than the perceptron due to its probabilistic nature.
\item SVM to achieve the highest accuracy because it maximizes the margin and generalizes better.
\end{itemize}
\section{Related Work}

Bitkina et al. applied several machine learning algorithms that classified whether a person has good or poor sleep quality using more advanced actigraphy data. The algorithms include SVM, KNN, logistic regression, and Naive Bayes. The features their dataset utilized contained sleep duration, time in bed, number of awakenings, and awakening duration. This closely mirrors the dataset we use in our project. In the report of the study, they stated that SVM was among the best performing models, achieving classification accuracy between 80 and 85\%. This study's dataset was based on a sleep study involving 22 male participants aged 22-40, while our mock dataset contains more than a thousand mock samples of a wider age and gender range. Their model found that time in bed and the time to awakening in minutes had the strongest correlations with sleep quality. Our algorithm toolset dives into linear classifiers and their ability to predict sleep quality, along with the SVM approach. We are hoping that our algorithms can reach and possibly surpass the accuracy rate that this report surmounted.

\section{Methods}

\subsection{Data Pre-processing}

The dataset was loaded from a CSV file containing 1000 samples and multiple features related to sleep behavior and lifestyle habits. Several preprocessing steps were applied to ensure the data was suitable for machine learning models.

First, column names were cleaned by stripping whitespace to prevent indexing inconsistencies. The \texttt{ID} column was removed, as it does not contribute to prediction and serves only as a unique identifier.

The target variable, \texttt{SleepEfficiency}, was originally a continuous value between 0 and 1. To convert this into a binary classification problem, we applied a threshold of 0.85. Samples with sleep efficiency greater than or equal to 0.85 were labeled as 1 (good sleep), and all others were labeled as 0 (poor sleep).

Categorical variables were encoded into numerical form. Specifically, \texttt{Gender} was mapped to binary values (Male = 0, Female = 1), and \texttt{SmokingStatus} was mapped similarly (No = 0, Yes = 1).

Time-based features required additional preprocessing. The \texttt{Bedtime} feature was converted into seconds since midnight using datetime parsing. The \texttt{WakeupTime} feature contained inconsistent formats (including timestamps and duration-like strings), so a custom conversion function was implemented to standardize all values into seconds. Any values that could not be parsed were later removed.

After these transformations, all rows containing missing values were dropped to ensure model consistency.

We partitioned the data into training, validation, and testing sets using a 70/15/15 split. The training set was used to fit the models, the validation set for tuning, and the test set for final evaluation.

\subsection{Feature Selection}

To analyze the relevance of each feature, we computed Pearson correlation coefficients between each input feature and the binary sleep efficiency label. This provided insight into the linear relationships present in the dataset.

The analysis revealed that all features had very weak correlations with the target variable, with absolute values below 0.06. This indicates that no single feature strongly influences sleep efficiency in a linear manner.

As a result, no features were removed during preprocessing. Instead, all available features were kept to allow the models to potentially learn weak combined patterns across multiple variables.

\subsection{Machine Learning Models}

We evaluated three supervised classification models: linear perceptron, logistic regression, and support vector machine (SVM). All models were trained on the same preprocessed dataset for consistency.

\textbf{Linear Perceptron:}
The perceptron is a linear classifier that updates its weights iteratively when misclassifications occur. It attempts to find a separating hyperplane between the two classes. Due to its simplicity and lack of probabilistic output, it serves as a baseline model.

\textbf{Logistic Regression:}
Logistic regression models the probability of class membership using a sigmoid function. In our implementation, we used L2 regularization to prevent overfitting and set the maximum number of training iterations to 1000 to ensure convergence.

\textbf{Support Vector Machine (SVM):}
The SVM model aims to find the optimal hyperplane that maximizes the margin between classes. A linear kernel was used to maintain consistency with the other models. The

regularization parameter C controls the trade-off between maximizing margin and minimizing classification error.

\subsection{Hyperparameter Choices}

For this stage of the project, most hyperparameters were kept at their default values to establish baseline performance. However, some key configurations were explicitly chosen:

\begin{itemize}
\item Logistic regression maximum iterations set to 1000 to ensure convergence
\item Class weighting enabled to handle dataset imbalance
\item Linear kernel used for SVM to maintain comparability across models
\end{itemize}

No extensive hyperparameter tuning was performed at this stage. Future work might include systematic tuning to improve performance.

\subsection{Evaluation Metrics}

Model performance was evaluated using four standard classification metrics:

\begin{itemize}
\item \textbf{Accuracy:} Measures overall correctness of predictions.
\item \textbf{Precision:} Measures how many predicted positive samples are actually correct.
\item \textbf{Recall:} Measures how many actual positive samples are correctly identified.
\item \textbf{F1 Score:} The harmonic mean of precision and recall, providing a balance between the two.
\end{itemize}

These metrics are particularly important due to the class imbalance in the dataset. Accuracy alone can be misleading, so precision, recall, and F1 score provide a more complete understanding of model performance.

\section{Preliminary Results}

\begin{table}[H]
\centering
\begin{tabular}{| c | c | c | c |}
\hline
& Linear Perceptron & Logistical Regression & Support Vector Model\\
\hline
Accuracy & 0.496 & 0.518 & 0.553 \\
\hline

```
        Precision & 0.230 & 0.184 & 0.193 \\
        \hline
        Recall & 0.560 & 0.391 & 0.345 \\
        \hline
        F1 & 0.326 & 0.250 & 0.247 \\
        \hline
    \end{tabular}
    \caption{Scores rounded to the nearest thousandth}
    \label{tab:placeholder}
\end{table}

\begin{table}[H]
  \centering
  \begin{tabular}{|l|r|}
    \hline
    \textbf{Feature} & \textbf{Correlation with Sleep Efficiency} \\
    \hline
     Age & -0.0523 \\
  \hline
  LightSleepPercentage & -0.0401 \\
  \hline
  DeepSleepPercentage & 0.0380 \\
  \hline
  ExerciseFrequency & 0.0376 \\
  \hline
  Awakenings & 0.0259 \\
  \hline
  SmokingStatus & 0.0174 \\
  \hline
  SleepDuration & -0.0131 \\
  \hline
  REMSleepPercentage & 0.0120 \\
  \hline
  CaffeineConsumption & 0.0107 \\
  \hline
  AlcoholConsumption & -0.0061 \\
  \hline
  Gender & -0.0010 \\
  \hline
  \end{tabular}
  \caption{Pearson correlation coefficients between each feature and Sleep Efficiency.}
  \label{tab:feature_correlation}
  \end{table}
```

\subsection{Results Discussion}

As shown in Figure 1, the preliminary results indicate that all three models perform at a near-random guessing level of 50\%, with accuracies ranging from 0.496 to 0.553. The support vector machine achieved the highest accuracy of 55.3\%, which proves that our proposal's expectation was correct. Despite this, the accuracy is still not strong enough to take seriously, and is not too marginally different from using the other linear classifier models that shared the value of not having predictive power.

One notable finding is that the linear perceptron model outperformed both logistic regression and SVM on recall (0.560) and F1-score (0.326). This is surprising given that it is the simplest model of the bunch. The linear perceptron updates its weights every time a sample is misclassified. Our dataset is quite imbalanced, with a significant lean towards poorer sleep quality samples (about 80\% of the dataset), meaning the update rule can cause the perceptron to change its decision boundary more freely. This comes at the expense of precision, where more good predictions were made on bad samples, leading to their low precision and accuracy values.

The uniformly low precision across the models can be attributed toward class imbalance of the dataset, where false positives originate from attempting to predict good sleep quality. The blame for poor performance can be shifted even more to the dataset, as analysis of feature correlation shows that there is little signal for the model to pick up on and apply to predicting the classifiers. In Figure 2, it is observed that no feature exceeds a correlation of 0.053. The strongest predictor was age, with a value of -0.052, indicating that older age is weakly associated with lower sleep efficiency. Features like AlcoholConsumption, CaffeineConsumption, and Awakenings proved to be noisy, as they showed essentially no linear relationship with the target classifier.

This is why our three machine learning algorithms failed in learning the classifier of the dataset. They hover around 50\% because it's the ceiling for a binary classification problem with no signals or clues to draw on. If the mock dataset had been constructed better or real data had been used, then this issue would have never originated and the accuracy and precision scores would be higher.

\section{Future plan}

Our future plans for this project may include searching for a better dataset that provides signaling features, potentially improving the model's accuracy. This would allow models to learn meaningful patterns rather than guess purely from noise. Lowering the threshold for good sleep quality may also be an appropriate choice to balance the dataset if we decide to keep the original one.

On the modeling side, we plan to conduct further hyperparameter tuning across all three classifiers. We may also experiment with other models we have learned in class content since beginning this project — KNN and Random Forest are contending options. Dimensionality reduction via PCA may also be applied to reduce the number of model features and confirm that the classes are separable.

\section{References}

Bitkina, Olga Vl et al. “Modeling Sleep Quality Depending on Objective Actigraphic Indicators Based on Machine Learning Methods.” International journal of environmental research and public health vol. 19,16 9890. 11 Aug. 2022, doi:10.3390/ijerph19169890

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